

Excellent! 100%

PHY201 Exam 2 on chapter(s): 4, 5, 6

Name: _____

These questions assess your understanding of the material from Chapter(s) 4, 5, 6 of OpenStax College Physics. Please write your name on the sheets of paper that you use. Carefully read each question. Credit is only awarded if you answer correctly and clearly show all major steps necessary to find your answer; credit is not awarded for missing work, unclear work, or the use of memorized equations absent from the equation sheet. Half credit may be awarded for insufficient work that suggests some degree of understanding. Half credit is awarded for correct numerical answers with incorrect or missing units. Correct numerical answers do not guarantee credit.

1. As Kyle waits for his food to heat up, he determines that his microwave oven rotates the food with a speed of 16.2 revolutions/minute. What is the food's angular velocity in radians per second?

$$16.2 \frac{\text{rev}}{\text{min}} \left| \frac{2\pi \text{ rad}}{1 \text{ rev}} \frac{\text{min}}{60 \text{ s}} \right| = 1.696 \frac{\text{rad}}{\text{s}}$$

ANSWER: $1.696 \frac{\text{rad}}{\text{s}}$

2. Abigail places an air-hockey puck on an air-hockey table that is inclined at an angle of 8.2° above the horizontal. The puck glides frictionlessly down the inclined surface. What is the puck's acceleration? The mass of the puck is 470 g and the mass of the air-hockey table is 40 kg.

ANSWER: $1.398 \frac{\text{m}}{\text{s}^2}$



$$W_{\parallel} = W \sin \theta = mg \sin \theta$$

$$\sum \vec{F}_{\parallel} = W_{\parallel} = ma$$

$$mg \sin \theta = ma$$

$$a = g \sin \theta = 9.8 \frac{\text{m}}{\text{s}^2} \cdot \sin 8.2^\circ = 1.398 \frac{\text{m}}{\text{s}^2}$$

3. Allison goes outside on a summer morning with her tea and sees a perfectly vertical strand of spider web. She watches for a while and eventually a spider climbs up the strand. At this point, invested in her experiment, Allison takes the spider and puts it on a scale, measuring its mass to be 405 mg, then returns it to its vertical strand of web. What is the tension in the vertical strand of web when the spider hangs from it?

$$405 \text{ mg} \left| \frac{1}{10^3} \frac{\text{g}}{\text{mg}} \frac{\text{kg}}{10^3 \text{ g}} \right| = 4.05 \times 10^{-4} \text{ kg}$$

ANSWER: $3.969 \times 10^{-3} \text{ N}$

$$\sum \vec{F}_y = 0$$

$$T = W$$

$$T = mg = 4.05 \times 10^{-4} \text{ kg} \cdot 9.8 \frac{\text{m}}{\text{s}^2} = 3.969 \times 10^{-3} \text{ N}$$

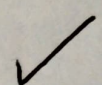
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4. Daniel and Andrew are playing rugby. Daniel pushes Andrew with a force of 1343 N. Daniel has a mass of 117 kg while Andrew has a mass of 91 kg. What is Andrew's acceleration as a result of getting hit by Daniel?

$$\vec{F} = m\vec{a}$$

$$\vec{a} = \frac{F}{m} = \frac{1343 \text{ N}}{91 \text{ kg}} = 14.758 \frac{\text{m}}{\text{s}^2}$$

ANSWER: 14.758 $\frac{\text{m}}{\text{s}^2}$



5. Allison is riding a motorcycle that can produce an acceleration of 3.1 m/s² while travelling at 70 km/h. The magnitude of the two forces resisting motion, friction and air resistance, total 690 N. The mass of Allison and the motorcycle is 390 kg. What is the magnitude of the force that the motorcycle's rear tire exerts on the ground?

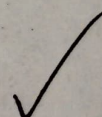
$$\Sigma \vec{F} = m\vec{a}$$

$$\Sigma \vec{F} = F_g - F_r$$

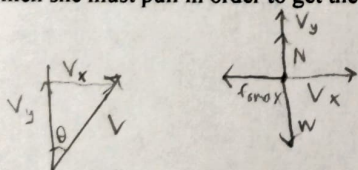
$$F_g - F_r = m\vec{a}$$

$$F_g = m\vec{a} + F_r = 390 \text{ kg} \cdot 3.1 \frac{\text{m}}{\text{s}^2} + 690 \text{ N} = 1899 \text{ N}$$

ANSWER: 1899 N



6. Victoria is a contestant in a winter sporting event. She is tasked with pulling a rope connected to a 410-kg block of ice on a frozen lake. Victoria pulls diagonally upward on the rope at an angle of 49° from the vertical. What is the minimum force with which she must pull in order to get the block of ice moving? Note that for ice on ice, $\mu_s = 0.1$ and $\mu_k = 0.03$.



ANSWER: 489.81 N

$$\Sigma \vec{F}_x = 0$$

$$f_{s \max} = F_{vx}$$

$$\mu_s N = F_v \sin \theta$$

$$\Sigma \vec{F}_y = 0$$

$$W = N + F_{vy}$$

$$N = mg - F_v \cos \theta$$

$$\mu_s (mg - F_v \cos \theta) = F_v \sin \theta$$

$$F_v \sin \theta + F_v \mu_s \cos \theta = \mu_s mg$$

$$F_v = \frac{\mu_s mg}{\sin \theta + \mu_s \cos \theta} = \frac{0.1 \cdot 410 \text{ kg} \cdot 9.8 \frac{\text{m}}{\text{s}^2}}{\sin 49^\circ + 0.1 \cos 49^\circ} = 489.81 \text{ N}$$



+3

7. Dylan is an astronaut walking on the surface of the planet Nova Terra. He experiences a small acceleration due to Earth's gravitational field, even at the distance between the center of Earth and Dylan of 0.29 Gm. What is his acceleration due to the force exerted by Earth's gravitational field alone?

$E 10^3$
 $M 10^6$
 $G 10^9$

$$F_g = G \frac{mM}{r^2}$$

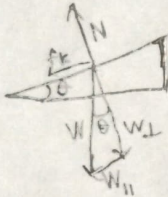
$$ma = G \frac{mM}{r^2}$$

$$F_g = ma$$

$$a = G \frac{M}{r^2} = 6.673 \times 10^{-11} \frac{\text{N} \cdot \text{m}^2}{\text{kg}^2} \cdot \frac{5.98 \times 10^{24} \text{ kg}}{(0.29 \times 10^9 \text{ m})^2} = 4.745 \times 10^{-3} \frac{\text{m}}{\text{s}^2}$$

ANSWER: $4.745 \times 10^{-3} \frac{\text{m}}{\text{s}^2}$

8. Nick is decelerating as he snowboards at 5.6 m/s up a slope that is inclined at 11.0° above the horizontal. Calculate the magnitude of his deceleration. The total mass of Nick and his snowboard is 92 kg. The density of the air is 1.278 kg/m^3 . Note that for waxed wood on snow, $\mu_s = 0.14$ and $\mu_k = 0.1$. Ignore the effect of air resistance.



$$\sum \vec{F}_\perp = 0$$

$$N = W_\perp$$

$$N = mg \cos \theta$$

$$\sum \vec{F}_\parallel = m\vec{a}$$

$$W_\parallel + f_k = ma$$

$$mg \sin \theta + \mu_k mg \cos \theta = ma$$

$$\vec{a} = g(\sin \theta + \mu_k \cos \theta)$$

$$\vec{a} = 9.8 \frac{\text{m}}{\text{s}^2} (\sin 11^\circ + 0.1 \cos 11^\circ) = 2.832 \frac{\text{m}}{\text{s}^2}$$

ANSWER: $2.832 \frac{\text{m}}{\text{s}^2}$

9. A 2.57-mg flea jumps into the air by exerting a force straight down on the ground. A breeze blows on the flying flea and exerts a horizontal force of 1 μN on the flea. Find the magnitude of the acceleration of the flea while it is airborne.

$$F = \sqrt{F_x^2 + F_y^2}$$

$$F_x = 1 \times 10^{-6} \text{ N}$$

ANSWER: $9.808 \frac{\text{m}}{\text{s}^2}$

$$F = ma$$

$$F_y = mg$$

$$ma = \sqrt{F_x^2 + F_y^2}$$

$$a = \frac{\sqrt{F_x^2 + F_y^2}}{m} = \frac{\sqrt{F_x^2 + (mg)^2}}{m} = \frac{\sqrt{(1 \times 10^{-6} \text{ N})^2 + (2.57 \times 10^{-6} \text{ kg} \cdot 9.8 \frac{\text{m}}{\text{s}^2})^2}}{2.57 \times 10^{-6} \text{ kg}} = 9.808 \frac{\text{m}}{\text{s}^2}$$

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1. 1.7 rad/s
2. 1.4 m/s²
3. 3.97 mN
4. 14.8 m/s²
5. 1899 N
6. 490 N
7. 4.74 mm/s²
8. 2.83 m/s²
9. 9.81 m/s²